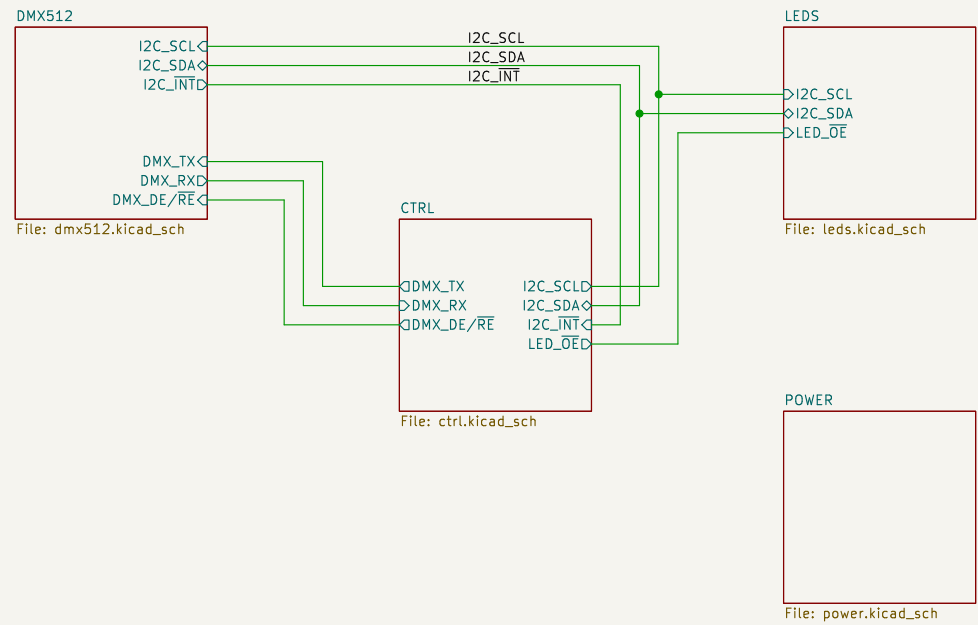


LED Beacon

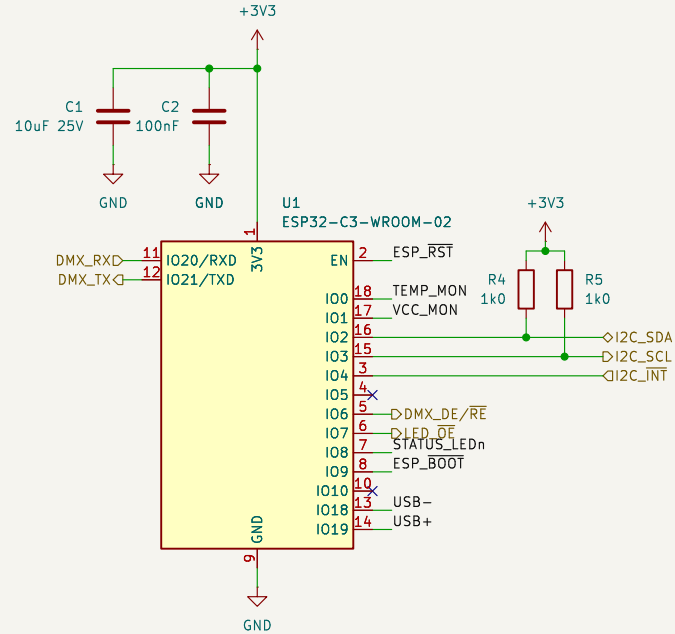
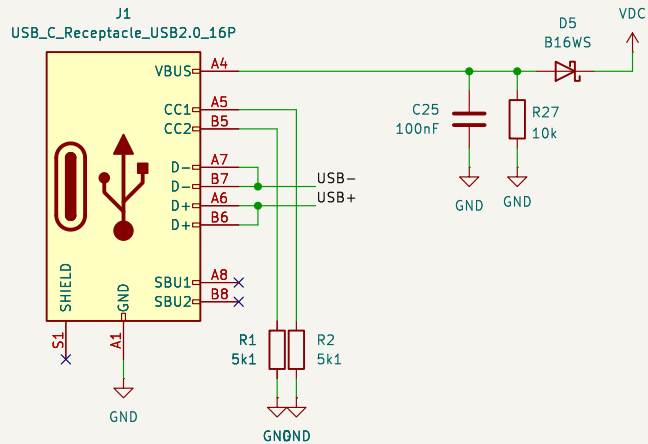


- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

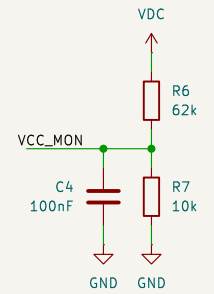
Wires & Bits		
Sheet: /		
File: emf_beacon.kicad_sch		
Title:		
Size: A4	Date: 2026-04-10	Rev: 0
KiCad E.D.A. 9.0.9		Id: 1/5

Controller

USB Programming

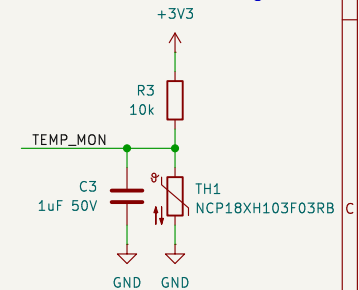


VCC Monitoring

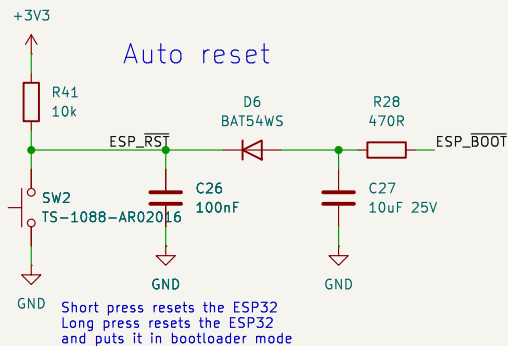


Voltage Monitor
 V_{max} is 17.5V
 $V_{adc} \times ((43k\Omega + 10k\Omega) + 10k\Omega) = VCC$
 $V_{adc} \times 5.3 = VCC$
 $1lsb = 4.271\mu V$

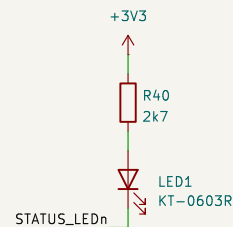
Temperature Monitoring



Auto reset



Status LED



I08 needs to be held high to boot
 so LED & resistor are high side.
 Set STATUS_LEDn to 0V to light LED.

Wires & Bits

Sheet: /CTRL/
 File: ctrl.kicad_sch

Title:

Size: A4
 Date: 2026-04-10
 KiCad E.D.A. 9.0.9

Rev: 0
 Id: 2/5

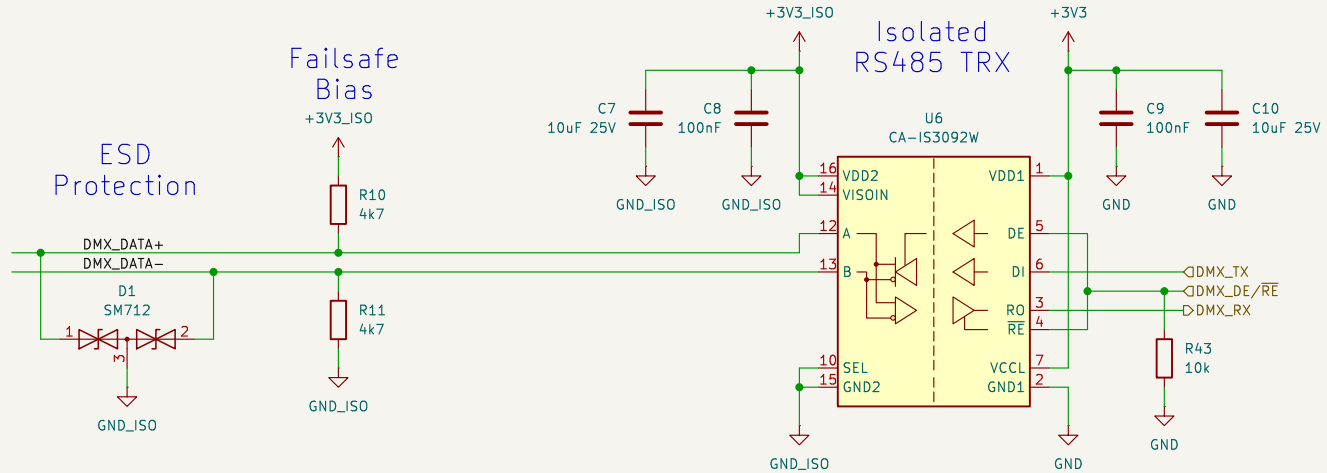
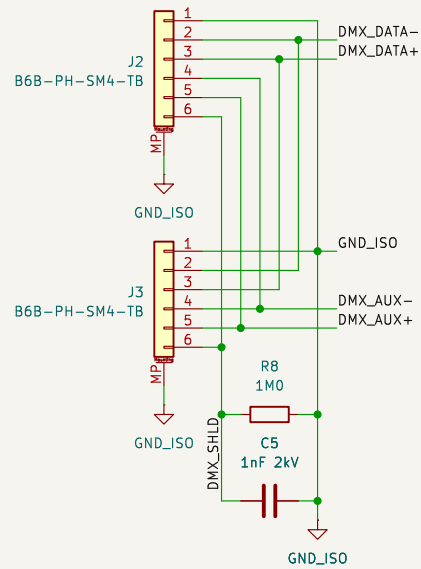
DMX512

Failsafe
Bias
+3V3_ISO

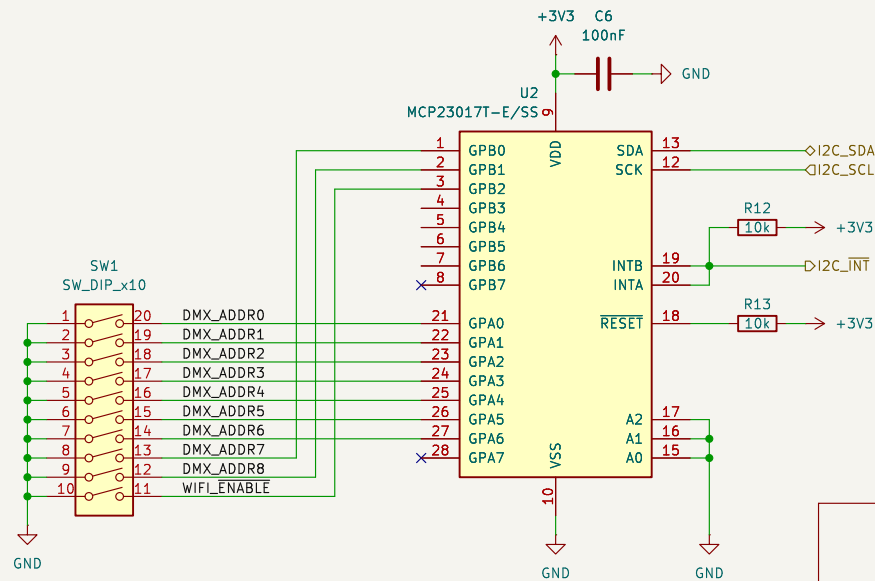
ESD
Protection

Isolated
RS485 TRX

To XLRs



CA-IS3092W has pin 14 as NC
but to maintain compatibility
with CA-IS3092VW it's
connected to 3V3_ISO



I²C Addr: 0x20
MCP23017 has a fault with GPA7 & GPB7.
The can corrupt an I²C transaction if
they change state during it.

Wires & Bits

Sheet: /DMX512/
File: dmx512.kicad_sch

Title:

Size: A4
KiCad E.D.A. 9.0.9

Date: 2026-04-10

Rev: 0

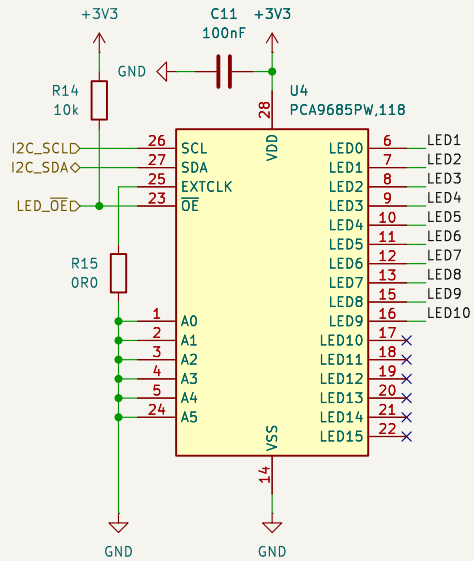
Id: 3/5

LED Driver

LED Enable

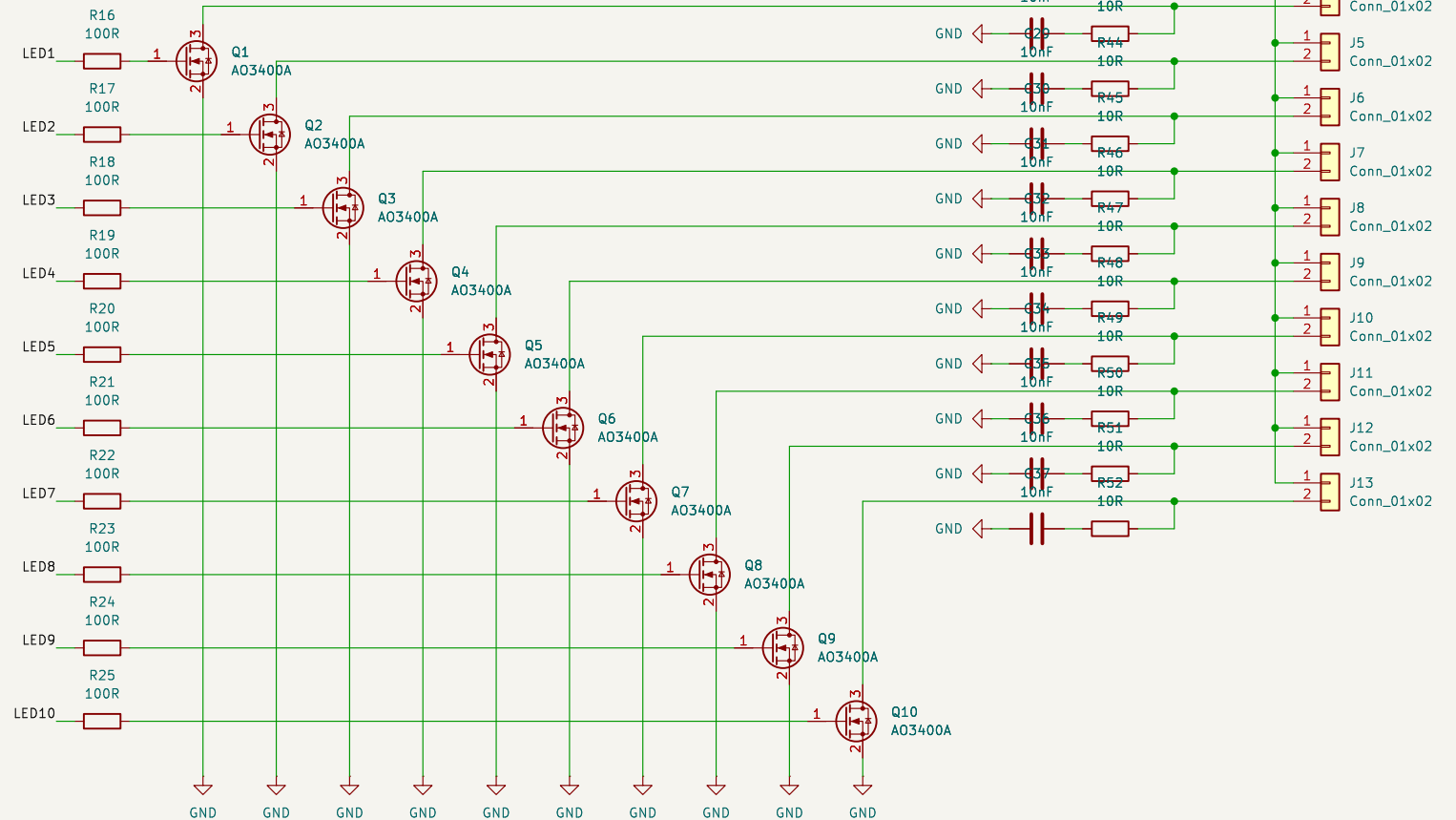
LED Sticks

I2C → PWM



I2C Addr: 0x40

A string of 6x LEDs is 6.4Vf @ 0.3A



Wires & Bits

Sheet: /LEDS/
File: leds.kicad_sch

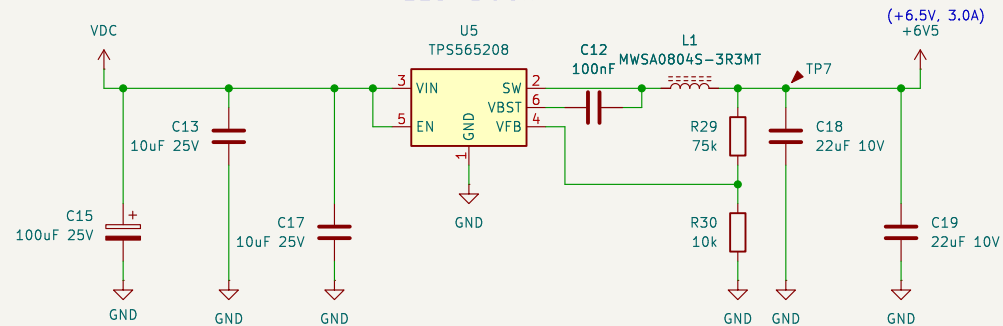
Title:

Size: A4 Date: 2026-04-10
KiCad E.D.A. 9.0.9

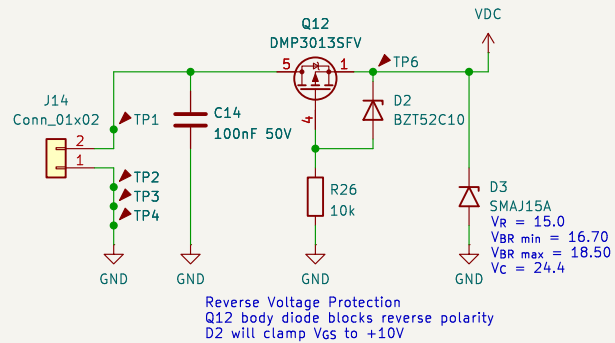
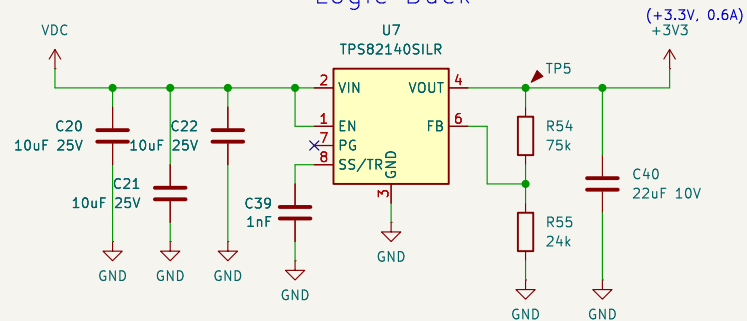
Rev: 0
Id: 4/5

Power Supplies

LED Buck



Logic Buck



EMC ferrites /CMC?

Wires & Bits

Sheet: /POWER/
File: power.kicad_sch

Title:

Size: A4
Date: 2026-04-10
KiCad E.D.A. 9.0.9

Rev: 0
Id: 5/5